Classification Model Performance – Questions and Answers

1. Accuracy

Question:

A classification model tested 100 samples. It correctly classified 85 samples.

Questions:

2. How many samples were classified incorrectly?

3. Calculate the accuracy of the model.

4. Calculate the error rate.

5. Is the model performing well based on accuracy?

6. What happens to accuracy if 5 more samples are classified correctly?

Ans:

2. How many samples were classified incorrectly?

Incorrect = Total − Correct

= 100 − 85

= 15

Answer: 15 samples

3. Calculate the accuracy of the model.

Accuracy = (Correctly Classified / Total Samples) × 100

= (85 / 100) × 100

= 85%

Answer: 85% accuracy

4. Calculate the error rate.

Error Rate = (Incorrectly Classified / Total Samples) × 100

= (15 / 100) × 100

= 15%

Answer: 15% error rate

5. Is the model performing well based on accuracy?

Yes. The model has 85% accuracy, meaning it correctly classifies 85 out of every 100 samples.

Answer: The model is performing reasonably well based on accuracy.

6. What happens to accuracy if 5 more samples are classified correctly?

New correct classifications = 85 + 5 = 90

Assuming the total number of samples remains 100:

New Accuracy = (90 / 100) × 100

= 90%

Answer: Accuracy increases from 85% to 90%.

2. Confusion Matrix

Question:

A model gives the following results:

- **True Positive (TP) = 40**
- **True Negative (TN) = 35**
- **False Positive (FP) = 10**
- **False Negative (FN) = 15**

Questions:

1. What is the total number of predictions?

2. Calculate the accuracy.

3. Calculate the error rate.

4. Which value represents incorrectly predicted positive cases?

5. Which value represents incorrectly predicted negative cases?

1. What is the total number of predictions?

Total Predictions = TP + TN + FP + FN

= 40 + 35 + 10 + 15

= 100

Answer: 100 predictions

2. Calculate the accuracy.

Accuracy = (TP + TN) / (TP + TN + FP + FN) × 100

= (40 + 35) / 100 × 100

= 75%

Answer: 75%

3. Calculate the error rate.

Error Rate = (FP + FN) / Total × 100

= (10 + 15) / 100 × 100

= 25%

Answer: 25%

4. Which value represents incorrectly predicted positive cases?

False Positive (FP) = 10.

A False Positive occurs when the actual class is negative, but the model predicts positive.

Answer: FP = 10

5. Which value represents incorrectly predicted negative cases?

False Negative (FN) = 15.

A False Negative occurs when the actual class is positive, but the model predicts negative.

Answer: FN = 15

3. Precision and Recall

Question:

A disease prediction system produces:

- **TP = 80**
- **FP = 20**
- **FN = 10**
- **TN = 90**

Questions:

1. Calculate Precision.

2. Calculate Recall.

3. Which metric tells us how many predicted positive cases were actually positive?

4. Which metric tells us how many actual positive cases were correctly identified?
5. Which metric is more important when missing a positive case is dangerous?

1. Calculate Precision.

Precision = TP / (TP + FP) × 100

= 80 / (80 + 20) × 100

= 80%

Answer: Precision = 80%

2. Calculate Recall.

Recall = TP / (TP + FN) × 100

= 80 / (80 + 10) × 100

= 80 / 90 × 100

= 88.89%

Answer: Recall = 88.89%

3. Which metric tells us how many predicted positive cases were actually positive?

The answer is Precision.

Precision tells us, of all cases predicted as positive, how many were actually positive.

Answer: Precision

4. Which metric tells us how many actual positive cases were correctly identified?

The answer is Recall.

Recall tells us, of all actual positive cases, how many were correctly identified.

Answer: Recall

5. Which metric is more important when missing a positive case is dangerous?

The answer is Recall.

Recall is important because it measures how many actual positive cases are correctly identified and helps minimize False Negatives.

Answer: Recall

4. Compare Two Classifiers

Question:

Two classifiers are tested on the same dataset.

Metric | Classifier A | Classifier B

Accuracy | 90% | 85%

Precision | 80% | 95%

Recall | 70% | 90%

Questions:

1. Which classifier has higher accuracy?

2. Which classifier has higher precision?

3. Which classifier has higher recall?

4. Which classifier would you choose if detecting as many positive cases as possible is important?

5. Give one reason why accuracy alone may not be enough.

1. Which classifier has higher accuracy?

Classifier A = 90%, Classifier B = 85%.

Answer: Classifier A has higher accuracy (90%).

2. Which classifier has higher precision?

Classifier A = 80%, Classifier B = 95%.

Answer: Classifier B has higher precision (95%).

3. Which classifier has higher recall?

Classifier A = 70%, Classifier B = 90%.

Answer: Classifier B has higher recall (90%).

4. Which classifier would you choose if detecting as many positive cases as possible is important?

We should choose the classifier with higher recall because recall measures how many actual positive cases are correctly identified.

Classifier A recall = 70%; Classifier B recall = 90%.

Answer: Classifier B, because it has 90% recall.

5. Give one reason why accuracy alone may not be enough.

Accuracy can be misleading when the dataset is imbalanced.

For example, if 95 out of 100 cases are negative, a model that predicts every case as negative can have 95% accuracy but detect none of the positive cases.

Answer: Accuracy alone may be misleading for imbalanced datasets; precision and recall should also be considered.

5. Analyze Model Performance

Question:

A classification model produces the following results:

- **TP = 70**
- **TN = 80**
- **FP = 20**
- **FN = 30**

Questions:

1. Calculate accuracy.

2. Calculate precision.

3. Calculate recall.

4. Calculate the error rate.

5. Based on these values, identify one strength and one weakness of the model.

1. Calculate accuracy.

Total = TP + TN + FP + FN

= 70 + 80 + 20 + 30 = 200

Accuracy = (TP + TN) / Total × 100

= (70 + 80) / 200 × 100

= 75%

Answer: Accuracy = 75%

2. Calculate precision.

Precision = TP / (TP + FP) × 100

= 70 / (70 + 20) × 100

= 70 / 90 × 100

= 77.78%

Answer: Precision = 77.78%

3. Calculate recall.

Recall = TP / (TP + FN) × 100

= 70 / (70 + 30) × 100

= 70%

Answer: Recall = 70%

4. Calculate the error rate.

Error Rate = (FP + FN) / Total × 100

= (20 + 30) / 200 × 100

= 25%

Answer: Error Rate = 25%

Verification: 100% − 75% = 25%

5. Based on these values, identify one strength and one weakness of the model.

Strength: The model has a precision of 77.78%, meaning most cases predicted as positive are actually positive. Its 75% accuracy also shows reasonably good overall performance.

Weakness: The model has a recall of only 70%, meaning it misses 30% of the actual positive cases.

Answer: Strength – good precision (77.78%). Weakness – recall is only 70%.

Topic 2. Prediction Algorithms – Questions and Answers

1. Simple Linear Prediction

Question:

A company's sales are predicted based on advertising expenditure.

Advertising Cost (₹)	Sales
10	20
20	40
30	60
40	80

Assume the relationship is linear.

Questions:

1. Identify the relationship between advertising cost and sales.

2. Predict sales when advertising cost is ₹50.

3. What happens to sales when advertising cost increases by ₹10?

4. Is this an example of classification or prediction?

5. Why is a linear prediction algorithm suitable here?

1. Identify the relationship between advertising cost and sales.

The relationship is directly linear.

From the table, every ₹10 increase in advertising cost increases sales by 20 units.

Therefore, the relationship can be written as:

Sales = 2 × Advertising Cost

Answer: Sales increase proportionally with advertising cost.

2. Predict sales when advertising cost is ₹50.

Sales = 2 × Advertising Cost

= 2 × 50

= 100

Answer: Predicted sales = 100 units.

3. What happens to sales when advertising cost increases by ₹10?

Every ₹10 increase in advertising cost results in a 20-unit increase in sales.

Answer: Sales increase by 20 units.

4. Is this an example of classification or prediction?

The output is a numerical value (sales).

Answer: This is a prediction problem, specifically regression.

5. Why is a linear prediction algorithm suitable here?

The data shows a constant rate of change: every ₹10 increase in advertising cost produces a 20-unit increase in sales.

Therefore, the data follows a straight-line relationship.

Answer: Linear prediction is suitable because the relationship between advertising cost and sales is approximately linear.

2. Predict Temperature

Given:

Day	Temperature (°C)
1	25
2	26
3	27
4	28
5	?

6	**?**
7	**?**
8	**?**
9	**?**
10	**?**
11	**?**
12	**?**
13	**?**
14	**?**
15	**?**

We can see that the temperature increases by 1°C every day.

1. Predict the temperature on Day 5.

Day 4 = 28°C
28+1= : 29°C
Answer: 29°C

2. Predict the temperature on Day 10.

Since the temperature increases by 1°C each day:

25+(10-1)(1)

= 25 + 9 = 34°C

Answer: 34°C

3. What is the predicted temperature on Day 15?

25+(15-1)(1)

= 25 + 14= 39°C

Answer: 39°C

4. How much does the temperature increase for every additional day?

From the given data:

- Day 1 ⟶ 25°C
- Day 2 ⟶ 26°C
- Day 3 ⟶ 27°C
- Day 4 ⟶ 28°C

Therefore:
26-25= 1°C

Answer: The temperature increases by 1°C for every additional day.

5. Is this a classification or prediction problem?

The output is a numerical temperature value, not a category.

Answer: This is a prediction problem, specifically numerical prediction (regression).

Final Answers

Question	Answer
Day 5	29°C
Day 10	34°C
Day 15	39°C
Increase per day	1°C
Problem type	Prediction / Regression

3. House Price Prediction

Question:

A prediction algorithm estimates house prices based on area.

Area (sq.ft)	Price (₹ lakhs)
1000	40
1500	60
2000	80

Assume a linear relationship.

Questions:
1. Identify the relationship between area and price.
2. Predict the price of a 2500 sq.ft house.
3. What happens to the price when area increases by 500 sq.ft?
4. Which type of algorithm can be used for this problem?
5. Is this classification or numerical prediction?

1. Identify the relationship between area and price.

The relationship is linear and directly proportional.

Every additional 500 sq.ft increases the price by ₹20 lakhs.

Answer: House price increases linearly as area increases.

2. Predict the price of a 2500 sq.ft house.

From the data, 1000 sq.ft → ₹40 lakhs and every 500 sq.ft adds ₹20 lakhs.

For 2500 sq.ft:

Price = ₹40 + [(2500 − 1000) / 500 × ₹20]

= ₹40 + (1500 / 500 × ₹20)

= ₹40 + ₹60

= ₹100 lakhs

Answer: Predicted price = ₹100 lakhs (₹1 crore).

3. What happens to the price when the area increases by 500 sq.ft?

From the table:

1500 − 1000 = 500 sq.ft, while 60 − 40 = ₹20 lakhs.

Answer: Price increases by ₹20 lakhs for every additional 500 sq.ft.

4. Which type of algorithm can be used for this problem?

A Linear Regression algorithm can be used because the relationship between area and price is linear.

Answer: Linear Regression.

5. Is this classification or numerical prediction?

The output is a numerical house price.

Answer: This is numerical prediction, specifically a regression problem.

4. Compare Actual and Predicted Values

Question:

A prediction algorithm gives the following results:

Actual Value	Predicted Value
50	48
60	65
70	68
80	85
90	88

Questions:

1. **Which prediction has the smallest error?**
2. **Which prediction has the largest error?**
3. **Calculate the error for each prediction.**
4. **Is the model generally predicting close to the actual values?**
5. **Why is the difference between actual and predicted values important?**

1. Which prediction has the smallest error?

The absolute errors are 2, 5, 2, 5, and 2.

The smallest error is 2.

Answer: Predictions for 50 → 48, 70 → 68, and 90 → 88 have the smallest error of 2.

2. Which prediction has the largest error?

The largest absolute error is 5.

Answer: Predictions for 60 → 65 and 80 → 85 have the largest error of 5.

3. Calculate the error for each prediction.

Error = Actual Value − Predicted Value.

$50 - 48 = 2$

$60 - 65 = -5$

$70 - 68 = 2$

$80 - 85 = -5$

90 − 88 = 2

Therefore, the absolute errors are: 2, 5, 2, 5, 2.

4. Is the model generally predicting close to the actual values?

Yes. All predictions are within 5 units of their actual values.

Answer: Yes, the model is generally predicting close to the actual values.

5. Why is the difference between actual and predicted values important?

The difference shows the prediction error of the model.

Smaller differences indicate that the model predictions are closer to the actual values and therefore more accurate.

Answer: It helps measure how well the prediction model is performing.

5. Choosing a Prediction Algorithm

Question:
Consider the following applications:
A. Predicting tomorrow's temperature.
B. Predicting the price of a house.
C. Predicting the number of products sold next month.
D. Predicting whether an email is spam or not.
Questions:
1. Which applications are prediction problems?
2. Which application is a classification problem?
3. Which algorithm could be used for predicting house prices?
4. Why is regression suitable for numerical prediction?
5. Give one difference between classification and prediction.

1. Which applications are prediction problems?

A. Predicting tomorrow's temperature.

B. Predicting the price of a house.

C. Predicting the number of products sold next month.

Answer: A, B, and C are numerical prediction/regression problems.

2. Which application is a classification problem?

D. Predicting whether an email is spam or not.

The output has categories: Spam or Not Spam.

Answer: D is a classification problem.

3. Which algorithm could be used for predicting house prices?

Linear Regression can be used when house price has a roughly linear relationship with features such as area.

Answer: Linear Regression.

4. Why is regression suitable for numerical prediction?

Regression algorithms are designed to predict continuous numerical values such as temperature, price, sales, and revenue.

Answer: Regression is suitable because it predicts numerical/continuous outputs.

5. Give one difference between classification and prediction.

Classification predicts a category or class, such as Spam or Not Spam.

Numerical prediction (regression) predicts a numerical value, such as ₹100 lakhs or 35°C.

Answer: Classification predicts categories, while regression predicts numerical values.